Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam A · Form 1: Foundations

Multiple Choice Questions: Select the best answer. (1 point each)

1. From the same tabletop simultaneously drop A and launch B horizontally. Ignore drag. Which arrives first?

(A) B

(B) A

(C) They arrive together.

(D) The heavier ball

2. A robot moves 6 m east, 8 m north. What is the magnitude of its displacement?

(A) -10 m

(B) 20 m

(C) 5 m

(D) 10 m

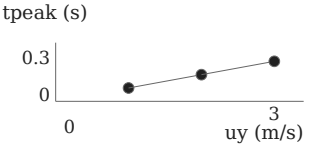
3. In gravity g=10 m/s² vary the upward component of launch velocity and time the ascent. uy (m/s) -> tpeak (s): (1, 0.1); (2, 0.2); (3, 0.3). Which vertical reading does the model predict at zero horizontal coordinate?

(A) 0.04

(B) -0.04

(C) 0.08

(D) 0



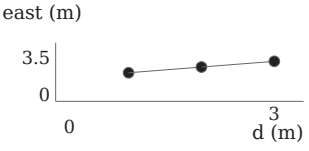
4. A map route starts at east coordinate 2 m and runs at 60° north of east. d (m) -> east (m): (1, 2.5); (2, 3); (3, 3.5). What is the signed change in the vertical reading from first to last trial?

(A) 1.4

(B) 0.6

(C) 1

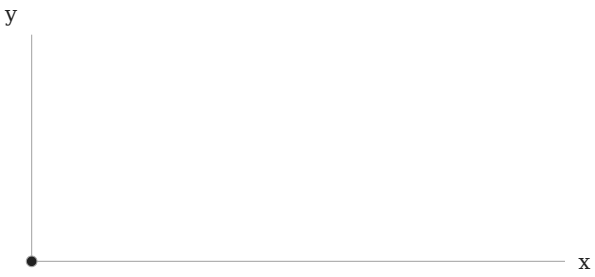
(D) 1.8



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A ball is launched from ground at 12 m/s, 20° above horizontal, and lands on the same ground. Ignore drag.

a. Draw the initial velocity vector and its horizontal and vertical components. **(1 pt)**



b. Find flight time, range, and velocity and acceleration at the top. **(3 pt)**

6. A ball launches horizontally from a 1.25 m table. Use g=10 m/s² and neglect drag. vx (m/s) -> R (m): (1, 0.5); (2, 1); (3, 1.5).

a. Build the model for horizontal-launch target from physics and predict both slope and intercept. **(2 pt)**

b. Plot the first and last measurements and describe the trend. **(2 pt)**

7. A ball is launched horizontally at speed u from height h without drag. Derive flight time and range R. **(3 pt)**